

HIRING PROCESS ANALYSIS

PROJECT DESCRIPTION: The project was designed to conduct data analytics on the hiring process within a company, with the primary objective of gaining insights into various aspects such as gender distribution, salary analysis, departmental composition, and position tiers. By analyzing these factors with the desired visualization charts the project aimed to provide valuable insights into hiring patterns

APPROACH: To perform the project a systematic approach was followed. A dataset having relevant information on hires, including gender, salary, department, and position, was obtained. Microsoft Excel 2021 was selected as the primary tool for data analysis due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, and formulas were employed to analyze the dataset and extract meaningful insights.

TECH-STACK USED: Software: Microsoft Excel 2021

Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the hiring data

INSIGHTS: various key insights were uncovered through the data analytics process. Meaningful trends and patterns were observed in the data

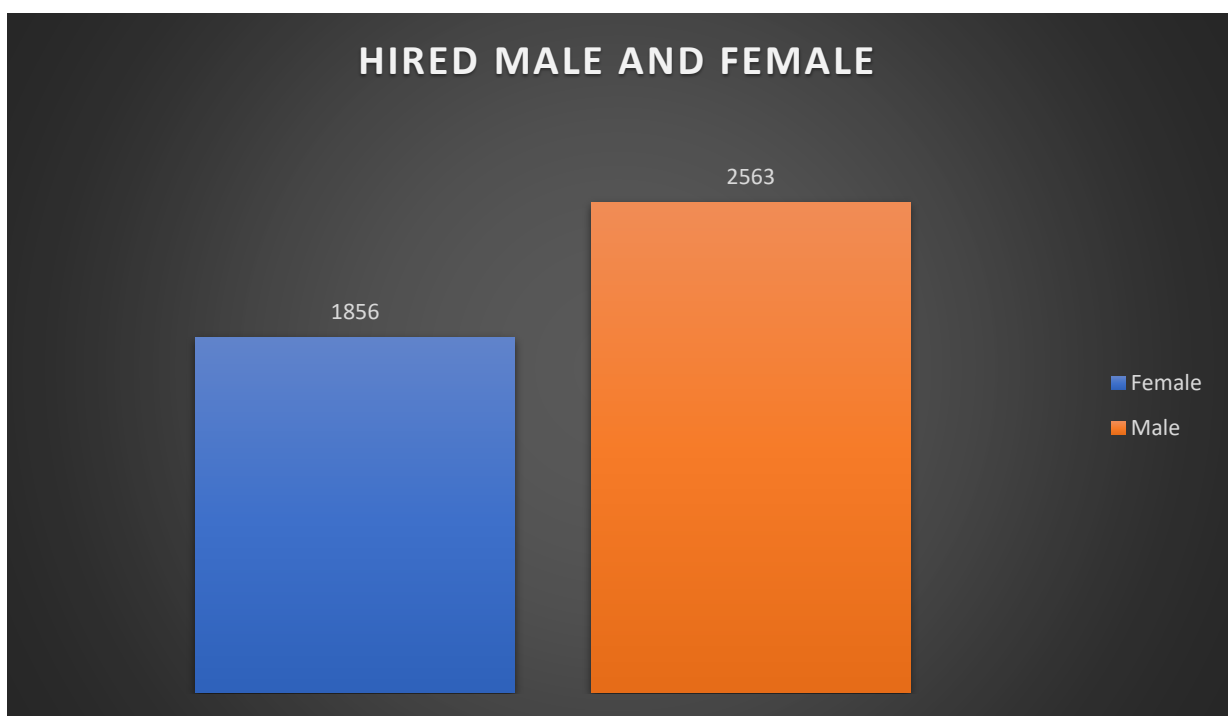
- Analysis of gender distribution
- Salary analysis revealed.
- Departmental composition
- Analysis of position tiers

TASKS:

A. Hiring Analysis: The hiring process involves bringing new individuals into the organization for various roles.

Your Task: Determine the gender distribution of hires. How many males and females have been hired by the company?

Count of event_name	event_name		
Status	Female	Male	Grand Total
Hired	1856	2563	4419
Grand Total	1856	2563	4419

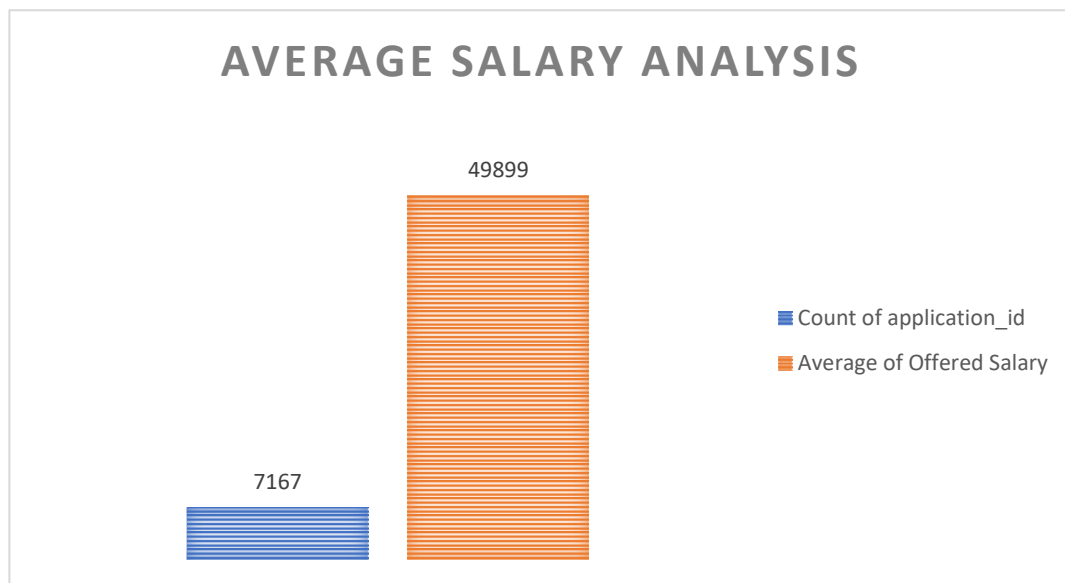


B. Salary Analysis:

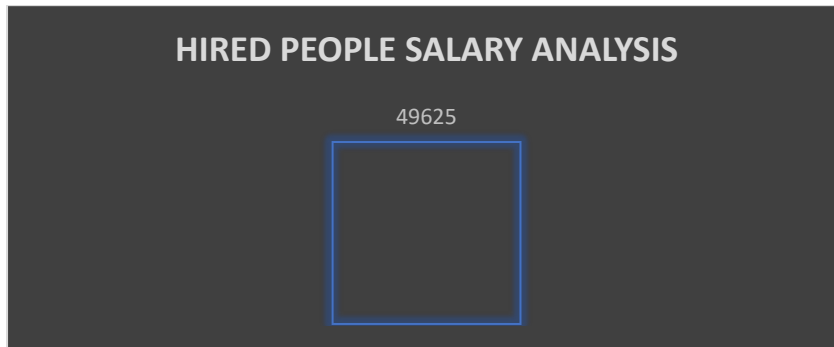
The average salary is calculated by adding up the salaries of a group of employees and then dividing the total by the number of employees.

Your Task: What is the average salary offered by this company? Use Excel functions to calculate this.

BY PIVOT TABLE	
Count of application_id	Average of Offered Salary
7167	49899



Status	Average of Offered Salary
Hired	49625
Grand Total	49625



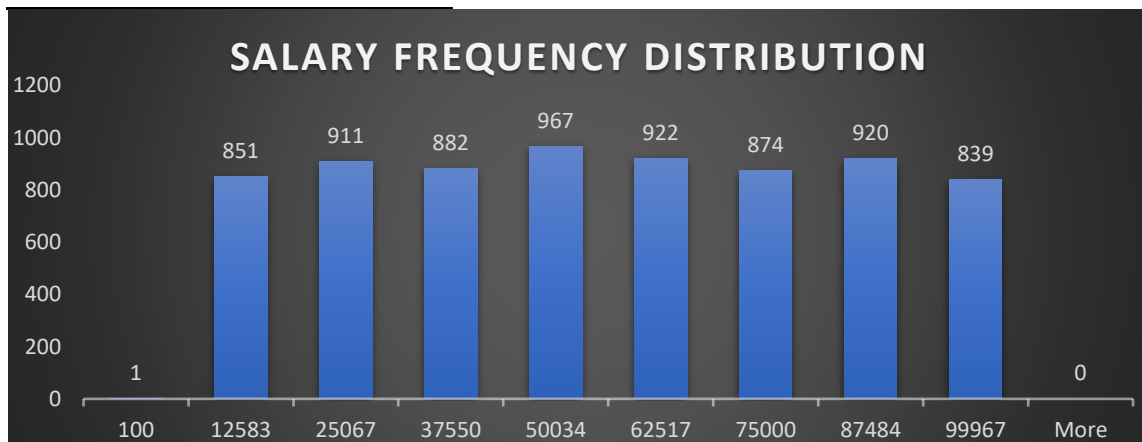
C. Salary Distribution:

Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

Your Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution.

Column1	
Mean	49899.3
Standard Error	335.0679
Median	49625
Mode	99967
Standard Deviation	28366.22
Sample Variance	8.05E+08
Kurtosis	-1.17945
Skewness	0.013242
Range	99867
Minimum	100
Maximum	99967
Sum	3.58E+08
Count	7167
99867	RANGE

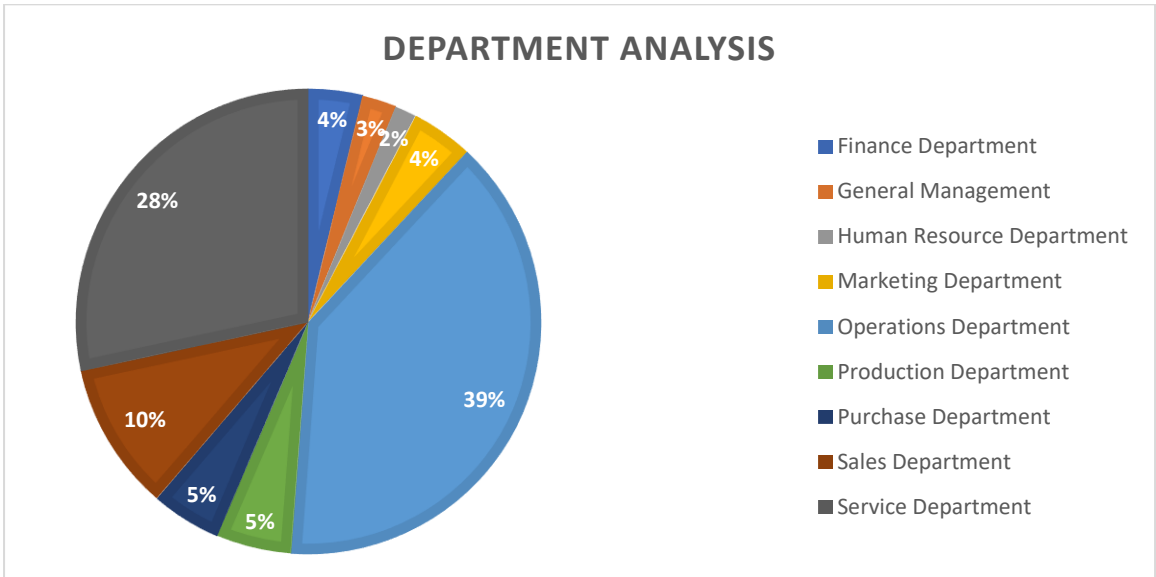
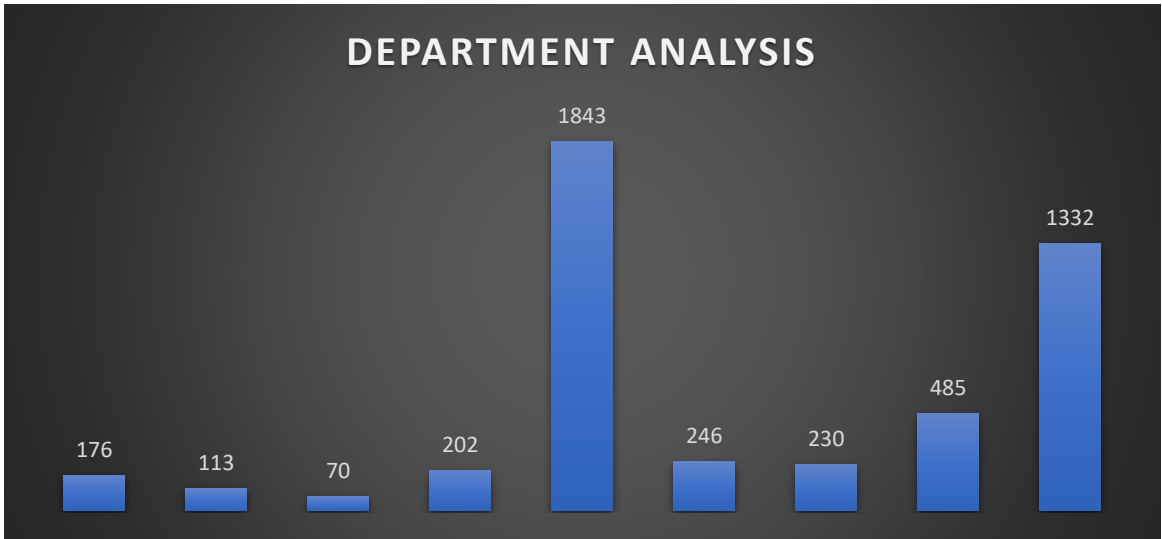
Bin	Frequency
100	1
12583	851
25067	911
37550	882
50034	967
62517	922
75000	874
87484	920
99967	839
More	0



D. Departmental Analysis: Visualizing data through charts and plots is a crucial part of data analysis.

Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

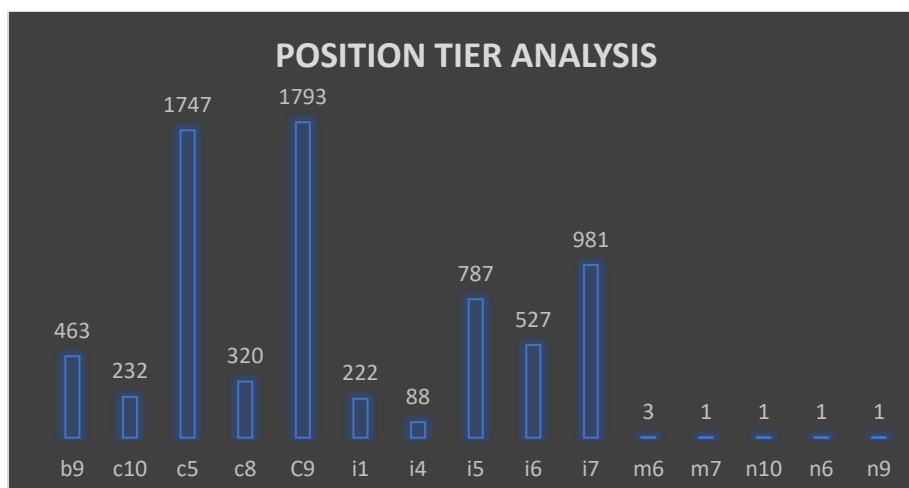
Count of Status	Status	
Department	Hired	Grand Total
Finance Department	176	176
General Management	113	113
Human Resource Department	70	70
Marketing Department	202	202
Operations Department	1843	1843
Production Department	246	246
Purchase Department	230	230
Sales Department	485	485
Service Department	1332	1332
Grand Total	4697	4697



E. Position Tier Analysis: Different positions within a company often have different tiers or levels.

Your Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Post Name	Count of Post Name
b9	463
c10	232
c5	1747
c8	320
C9	1793
i1	222
i4	88
i5	787
i6	527
i7	981
m6	3
m7	1
n10	1
n6	1
n9	1



RESULT:

Significant insights into the hiring process analytics and by analyzing hiring patterns, salary distribution, and departmental composition help optimize recruitment strategies and drive organizational growth

EXCEL REFER LINK:

[TRAINITY-PROJECT 4- HIRING DATA ANALYSIS.xlsx](#)

DATA CLEANING AND MANIPULATION SCREENSHOTS:\

114584	07-05-2014 08:08	Rejected	Male	Sales Department	i7	
	DELETING THE ROW BECAUSE IT IS A REJECTED CASE					

application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary
289907	01-05-2014 07:44	Hired	Male	Sales Department	C9	85914
955372	02-05-2014 09:16	Rejected	Female	Sales Department	b9	85176
882740	07-08-2014 13:42	Rejected	Male	Sales Department	c9	85837
230796	25-08-2014 09:32	Hired	Female	Sales Department	c9	85569
101190	09-05-2014 17:23	Hired	Male	Sales Department	c9	85057
804019	08-07-2014 16:52	Hired	Female	Sales Department	c9	85130
226229	10-07-2014 19:03	Hired	Male	Sales Department	c5	85610
788752	18-07-2014 17:37	Rejected	Male	Sales Department	c5	85392
294317	29-07-2014 17:33	Rejected	Female	Sales Department	b9	85400

Post Name			
c8			
c5			
c5			
i4		c-10 to c10	
i4			
C9			
i7			
i7			
i4			
i4			
n10			
b9			
b9			
b9			
i7			
i7			
i5			
i5			
i1			
i1			
i1			
i1			
i6			
i7			

[illegible]

Offered Salary
56553
22075
70069
3207
29668
85914
69904
11758
15156
49515
26990
200000
86787
2308
56688
81757
15134
100
73579
50351
38462
82510
52554
3423
88744
70979

OUTLIERS IN OFFERED SALARY

Box plot showing the distribution of Offered Salary. The plot includes a median line at 49,983.02, a mean line at 99,967, and a range from 2,545 to 74,456. Outliers are marked at 200,000, 300,000, and 400,000.

Legend: Offered Salary

Offered Salary
200000
400000
300000

WE CONSIDER THE OUTLIERS AND ASSIGN IT THE UPPER VALUE 99967

THANK YOU

SANGRAM KESHARI MOHAPATRA